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1. ABSTRACT

Academic underperformance and student dropout are problems in schools and universities. This is especially true in places where teachers and administrators do not have the time or resources to keep an eye on students. When students do not do well in school it affects not their own success but also the overall performance of the school. It can even have long-term effects on the students and society as a whole. Traditionally teachers and administrators have used things like exam scores and semester grades to identify students who are struggling. However these methods are not very effective because they only catch problems after the student has already fallen behind. With the help of technology and data we can now use machine learning to identify students who're at risk of underperforming before it is too late.

This study uses a kind of machine learning called Explainable Machine Learning to identify students who are at risk of doing poorly in school. We used a dataset that includes information about students grades, attendance, study habits and other factors. The goal of this study is not to identify students who are at risk but also to understand why they are at risk and to come up with personalized plans to help them succeed.

We used different machine learning models to predict which students were at risk including Logistic Regression, Decision Tree and XGBoost. We also used techniques to make the models more accurate, such as looking at attendance and quiz scores. The XGBoost model was accurate, correctly identifying at-risk students 92% of the time.

One of the problems with machine learning models is that they can be hard to understand. To solve this problem we used a technique called SHAP to explain why the model was making predictions. This helped us identify the important factors that affect academic risk, such as quiz scores and attendance. This makes it easier for teachers and administrators to understand why a student is at risk and to come up with plans to help them.

We also used a technique called anomaly detection to identify students who may be at risk even if their grades are okay. This helped us catch students who may be struggling in ways, such as with behavioral problems. 10% Of students were identified as being at risk in this way.

The results of this study show that using a combination of machine learning models, explainability and anomaly detection can help create an effective system for supporting students. This system can help answer questions like which students are at risk why they are at risk and what can be done to help them. This makes it a useful tool for schools and universities that want to improve student retention and success.

Feature selection was further studied compared to full feature model and reduced feature model based on XGBoost feature importance. The reduced model considered the most potent predictors, i.e., quiz average, attendance, failures, and performance trend.

On the other hand, full feature model was slightly better, which shows that additional academic, behavioral, and demographic features improved the quality of prediction via the interaction of features.

The framework we propose is highly suitable for real-world implementation in educational institutions. It can help improve student retention, academic success and term learning outcomes. The use of machine learning and data can help identify students who're at risk before it is too late.

By understanding why students are at risk teachers and administrators can come up with plans to help them succeed. The combination of modeling, explainability, anomaly detection and intervention planning makes for a powerful tool, in supporting students.

2. Introduction

Student academic underperformance and dropout are still problems in today's educational systems. This is especially true in schools where help's not readily available early on. Poor grades do not hurt a students chances of success. They also lead to dropouts, make schools less efficient and hurt students in the long run. Identifying students who are struggling is key to helping them stay in school and do better.

Usually teachers find out which students are struggling by looking at their exam results or by observing them.. These methods are too late. They only find out after the student has already fallen behind. With more and more data available we can use computers to predict which students might struggle. This way we can help them before its too late. Many researchers have used computer models to predict student performance. These models look at things like grades and behavior to guess if a student will do poorly.

Most of these models have two big problems. First they do not explain why they make the predictions they do. Second, they miss students who seem fine. Might be having hidden problems. Explainable AI is a way to make these models more transparent. It helps teachers understand why a student might be at risk. Anomaly detection can also help find students who seem okay. Are actually having issues. This research proposes a framework that uses computer models to predict academic risk. It also explains why a student might be at risk. Finds students who seem fine but might be having hidden problems. The framework uses models, including Logistic Regression, Decision Tree and XGBoost. It also uses SHAP to explain the predictions and Isolation Forest to find students.

The results show that XGBoost works best with an accuracy of 92%. The SHAP analysis found that quiz averages, attendance and prior failures are the important factors. The framework also developed a system that gives recommendations. This includes counseling, monitoring and reviewing behavior.

The main contribution of this research is a framework that predicts risk and explains why. It also supports interventions. This approach improves prediction accuracy and practicality. This research can help schools make decisions. It provides a practical and reliable way to detect academic risk early on. The proposed framework can be used in world educational decision support systems. It provides a layer of early warning for students who are at risk.

Another important challenge in predicting student performance is choosing the most relevant features for creating efficient and reliable models. Reducing the number of features can improve clarity and lower computational demands. However, cutting too many features may lead to losing key interactions among academic, behavioral, and demographic variables. Therefore, it is essential to compare full-feature models with reduced-feature models to find the right balance between prediction accuracy and model simplicity.

By using this framework schools can help more students succeed. They can also reduce dropout rates. Improve institutional efficiency. Early intervention is key to helping students who are struggling. This framework provides a way to identify students at risk and provide them with the help they need. It is a step towards improving educational outcomes. It can also help schools make use of their resources. The framework can be used to support students in educational settings. It provides a proactive approach to academic support.

By predicting risk early on, schools can provide targeted interventions. This can help students who are struggling and improve educational outcomes.

3. LITERATURE REVIEW

Predicting how well students will do in school and stopping them from dropping out has become a deal in education. With more and more students learning on computers and using systems to manage their schoolwork schools are getting a lot of data about their students. This data can be used to figure out if a student is having trouble on. Researchers have used methods to predict how well students will do and to help them stay in school. They want to make sure students get the help they need to succeed.

Chicon and his team made a model that could predict if a student was going to drop out. They used a few methods like looking at data and grouping students together to make their predictions. Their model was pretty accurate, getting it right 93.8% of the time. It also helped them understand what kinds of students were more likely to drop out and why.

Azcona and his team made a system that could predict if a student was going to drop out on a basis. They used information about the students. How they interacted with the online course material. They tried out a few methods and found that predicting dropouts every week helped them identify students who were struggling and get them help sooner. Kim and his team made a system that could predict if a student was going to drop out and why. They grouped the reasons into categories like having a job, personal issues or struggling in school. This helped the school give students the kind of counseling and support.

Usman and his team compared a few methods for predicting how well students would do. They used data from an online learning system. I found that one method, called Decision Tree, worked the best. They also found that whether or not a student turned in their assignments was an indicator of how well they would do in school. Some researchers are now focusing on making models that can explain why they make predictions. This is called Explainable Artificial Intelligence. It is helpful because it allows teachers to understand why a student is struggling and what they can do to help.

Some methods can even detect students who're at risk of dropping out but do not seem to be struggling. These methods look for behavior patterns that might indicate a problem. Even though researchers have made some progress in predicting student success most models only focus on making predictions. Do not explain why a student is at risk or what to do about it.

This study is trying to fix that by making a model that can predict student success, explain why a student is at risk, detect unusual behavior patterns and provide recommendations for what to do. This model is more useful for schools because it gives them a plan for how to help struggling students. The model doesn't just stop at predicting which students will drop out. It also helps teachers understand why a student is struggling and what they can do to help. This makes it more useful, for making decisions in the world.

4. METHODOLOGY

This research proposes an Explainable Machine Learning Framework for Early Academic Risk Detection and Intervention. The methodology combines supervised classification, explainable artificial intelligence, anomaly detection, and a rule-based recommendation system to identify academically at-risk students and support early intervention. The complete workflow consists of five major stages: data collection, data preprocessing, feature engineering, model development, and intervention generation.

4.1 Dataset Description

The dataset used in this study is the UCI Student Performance Dataset (student-mat.csv), obtained from the UCI Machine Learning Repository. The dataset contains academic, behavioral, and socio-demographic information of secondary school students and is widely used for student performance prediction research.

The dataset consists of 395 student records and 33 attributes before feature engineering. These attributes include academic scores, attendance patterns, family background, study habits, and social behavior indicators. The target of this study is the early identification of students at academic risk before final academic failure occurs.

The most relevant original features include:

- G1: First period grade
- G2: Second period grade
- failures: Number of past class failures
- studytime: Weekly study time
- absences: Number of school absences
- goout: Frequency of going out with friends
- internet: Internet access at home
- age: Student age

- Medu and Fedu: Mother's and father's education level
- famsize: Family size
- address: Urban or rural home location

These attributes were selected because they represent both academic performance and behavioral risk indicators.

Table 1: Dataset Features Used in the Study

Feature Type	Features
Academic Features	G1, G2, quiz_avg, failures
Behavioral Features	studytime, attendance, goout, internet
Demographic Features	age, Medu, Fedu, famsize, address
Engineered Features	attendance, quiz_avg, trend

4.2 Data Preprocessing

Before model training, the dataset was cleaned and transformed to improve prediction quality. Missing values and inconsistent records were checked and corrected to ensure data reliability. Categorical variables such as family size, address, and internet access were converted into numerical representations using label encoding.

The target variable called **risk** was created to classify students into two categories:

- Low Risk (0): Students expected to perform satisfactorily
- High Risk (1): Students likely to experience academic difficulty

This binary classification allows the machine learning models to perform early academic risk prediction rather than waiting for final academic failure.

Feature scaling and normalization were applied where necessary to improve model stability and reduce bias caused by varying feature ranges.

4.3 Feature Engineering

Feature engineering was performed to improve predictive performance and make the model more meaningful for real-world educational decision-making. Instead of relying only on raw dataset attributes, additional indicators were created to better represent student behavior and academic trends.

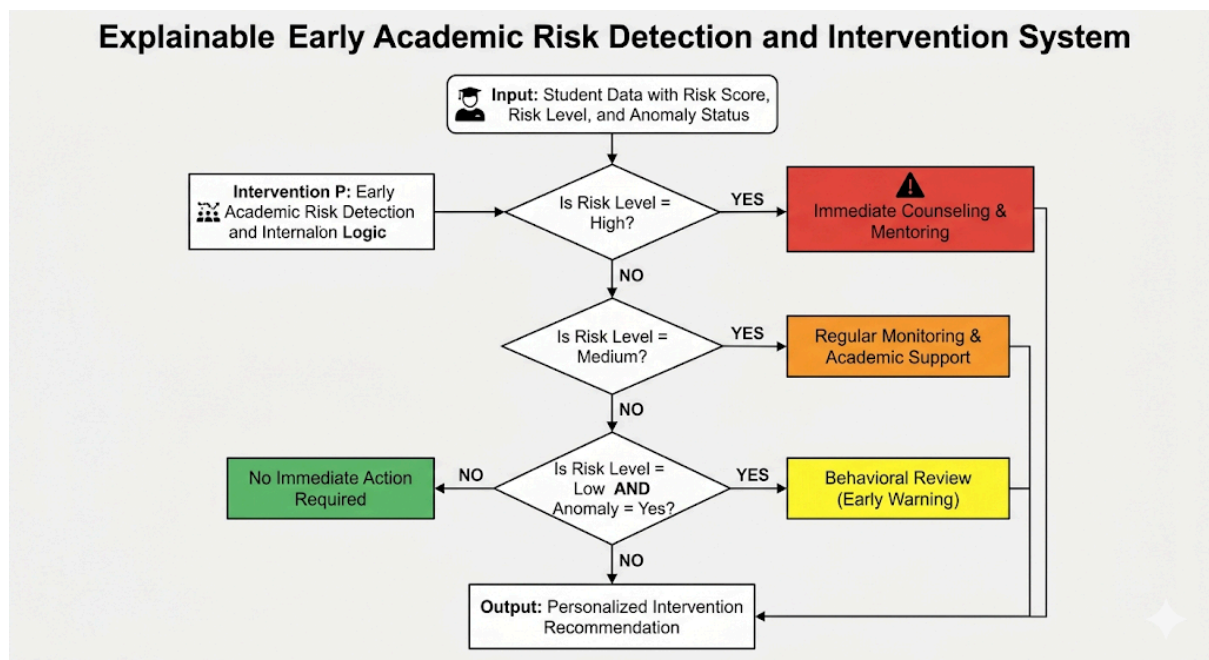


Fig 1: Proposed System Workflow for Early Academic Risk Detection and Intervention

The engineered features include:

Attendance

The original absences feature was transformed into an attendance indicator to better reflect classroom participation. Higher attendance generally indicates stronger academic engagement.

Quiz Average (quiz_avg)

A performance indicator called quiz average was created using:

$$\text{quiz_avg} = \frac{G1 + G2}{2}$$

This feature represents the student's average academic consistency across the first two grading periods.

Performance Trend (trend)

A performance trend feature was created using:

$$\text{trend} = G2 - G1$$

This feature helps identify whether student performance is improving or declining over time.

These engineered features significantly improved model interpretability and prediction quality.

4.4 Machine Learning Models

Three classification models were used for academic risk prediction:

Logistic Regression

Logistic Regression was used as the baseline model because of its simplicity and interpretability. It provides a strong reference point for comparing more advanced models.

Decision Tree

Decision Tree was used because of its transparent rule-based prediction process. It helps visualize academic risk factors and supports easier explanation for educators.

XGBoost

XGBoost was selected as the primary model because of its strong performance in handling structured tabular data and feature interactions. It also supports feature importance analysis and integrates well with SHAP explainability techniques.

The dataset was divided into training and testing sets to evaluate model performance fairly. Accuracy, Recall, Precision, F1-score, and Confusion Matrix were used as evaluation metrics.

4.5 Feature Selection Using XGBoost

Feature importance analysis was performed using XGBoost to identify the most influential variables affecting academic risk prediction. Important features such as quiz average, attendance, failures, and performance trend showed the strongest contribution to prediction accuracy.

A comparison was conducted between:

- Full Feature Model
- Reduced Feature Model

This comparison helped ensure that the final model maintained high performance while reducing unnecessary complexity.

To validate the feature selection further, a comparison was made between the full feature model and the reduced feature model. The reduced model included only the most important predictors, as identified by the XGBoost feature importance, whereas the full model included all academic, behavioural, and demographic variables. This comparison was to see if reducing feature complexity helped the model run better without losing much in prediction accuracy.

4.6 Explainable AI Using SHAP

To improve model transparency, SHAP (SHapley Additive Explanations) was used for explaining XGBoost predictions. SHAP helps identify how each feature contributes to an individual prediction and the overall model output.

This allowed the system to answer practical questions such as:

- Why is a student predicted to be high risk?
- Which academic or behavioral factors contributed most?
- What should educators focus on for intervention?

The most influential features identified were:

- quiz_avg
- attendance
- failures

This improves trust and usability for real-world academic decision-making.

4.7 Anomaly Detection Using Isolation Forest

Traditional classification models may miss students whose academic scores appear normal but whose behavioral patterns indicate future academic problems. To address this limitation, Isolation Forest was used for anomaly detection.

Approximately 10% of students were identified as anomalous. Some of these students were not classified as high risk by the prediction model but showed unusual behavioral patterns.

This finding is important because it provides an additional early warning mechanism for identifying hidden-risk students before visible academic decline occurs.

4.8 Rule-Based Intervention System

The final stage of the framework is a rule-based intervention system that generates personalized academic support recommendations using two factors:

- Risk prediction label
- Anomaly detection flag

Based on these conditions, the system recommends actions such as:

- Academic counseling
- Performance monitoring
- Behavioral review
- Personalized mentoring

This transforms the system from a predictive model into a practical educational decision support framework that can assist teachers and academic administrators in improving student retention and success.

5. RESULTS AND DISCUSSION

The proposed framework was evaluated using multiple machine learning models to identify students at academic risk and to determine the most effective approach for early intervention. The performance of Logistic Regression, Decision Tree, and XGBoost was compared using standard evaluation metrics including Accuracy, Recall, and F1-score. In addition, Explainable AI using SHAP and anomaly detection using Isolation Forest were analyzed to improve practical decision-making.

5.1 Model Performance Comparison

Three classification models were trained and tested on the processed dataset. Logistic Regression was used as the baseline model, Decision Tree provided interpretable rule-based predictions, and XGBoost served as the primary advanced model.

Table 2: Performance Comparison of Classification Models

The performance comparison is shown below:

Model	Accuracy	Recall	F1 Score
Logistic Regression	0.90	0.81	0.84

Decision Tree	0.84	0.73	0.75
XGBoost	0.92	0.88	0.88

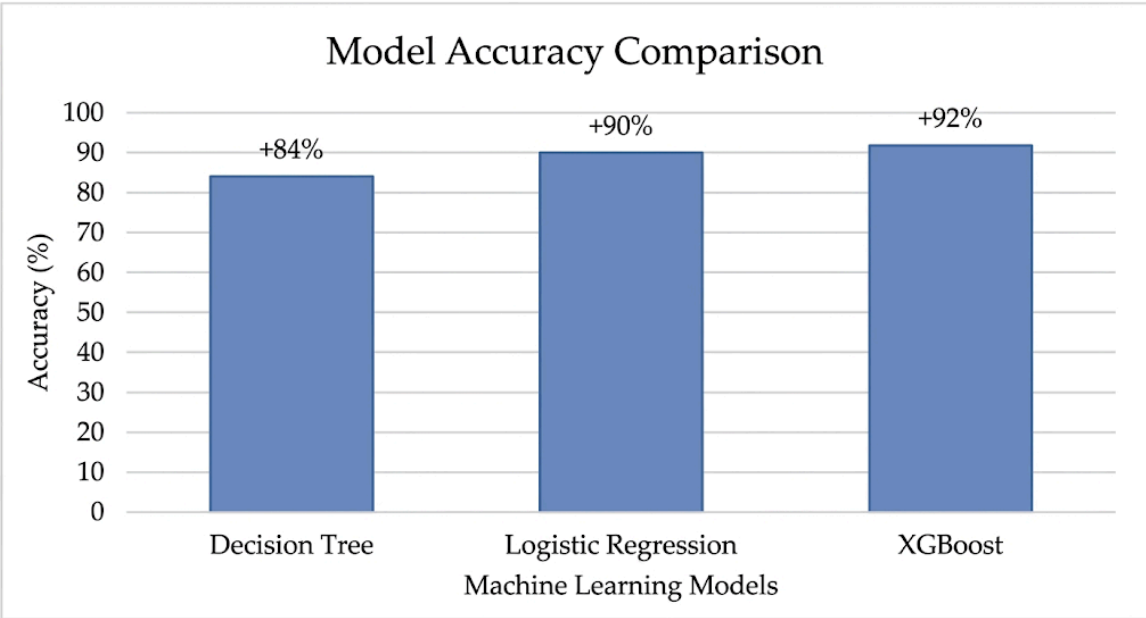


Fig 2: Accuracy Comparison of Classification Models

Among all models, XGBoost achieved the best performance with an accuracy of 92%, recall of 88%, and F1-score of 0.88. This indicates that XGBoost was more effective in correctly identifying high-risk students while minimizing false classifications.

Logistic Regression performed reasonably well and served as a reliable baseline, but it was less effective than XGBoost in handling complex feature relationships. Decision Tree showed lower performance, particularly in recall, which suggests that some high-risk students were not correctly identified.

The superior performance of XGBoost demonstrates that ensemble boosting methods are highly suitable for structured educational datasets where feature interactions significantly influence prediction outcomes.

5.2 Confusion Matrix Analysis

To further evaluate the best-performing model, the confusion matrix of XGBoost was analyzed.

Table 3: Confusion Matrix of XGBoost Model

Actual / Predicted	Low Risk (0)	High Risk (1)
Low Risk (0)	50	3
High Risk (1)	3	23

From the confusion matrix:

- True Negatives (TN): 50
- True Positives (TP): 23
- False Positives (FP): 3
- False Negatives (FN): 3

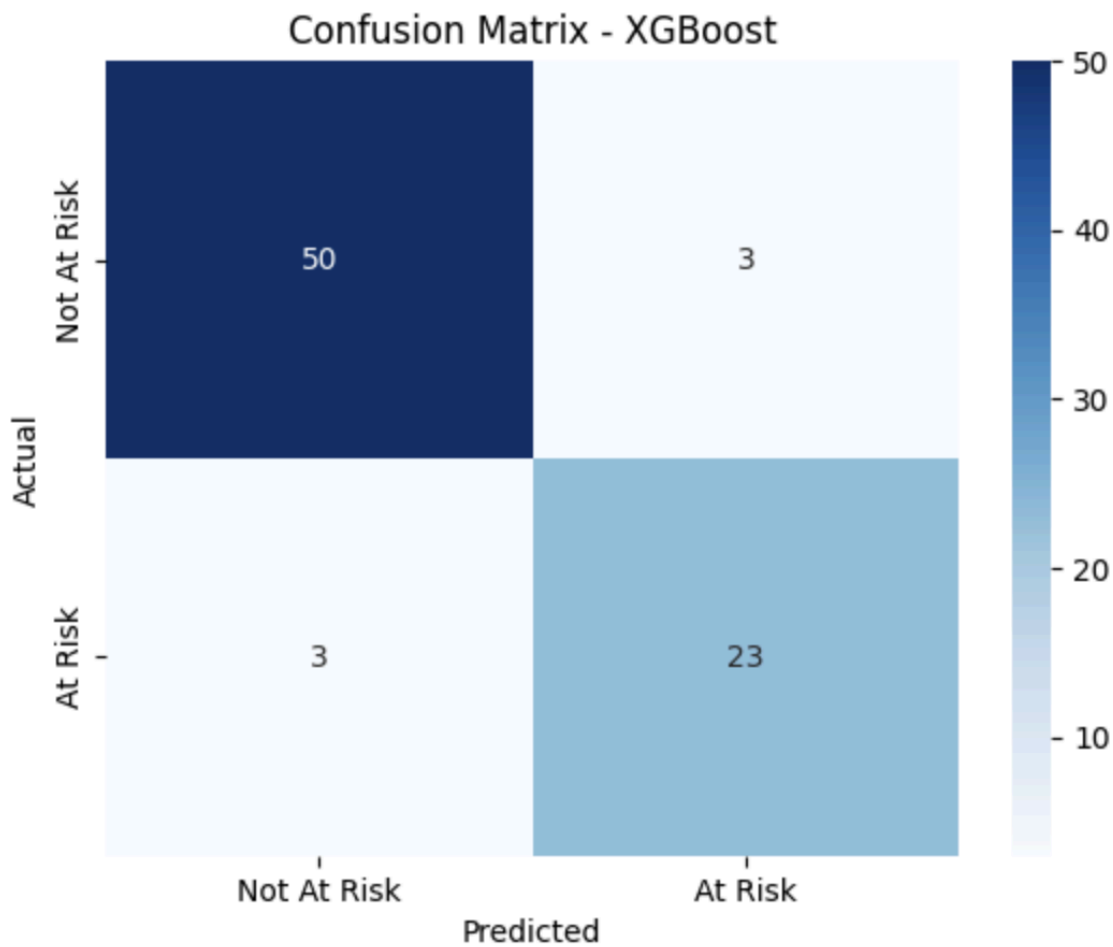


Fig 3: Confusion Matrix Heatmap of the XGBoost Model

The low number of false negatives is particularly important in educational prediction because missing a truly high-risk student can delay intervention and increase the probability of academic failure. Similarly, the low false positive count shows that unnecessary intervention for low-risk students is minimized.

This balance makes the model highly practical for real-world academic decision support systems.

5.3 Feature Importance and Selection

Feature importance analysis was performed using XGBoost to identify the strongest predictors of academic risk. The most influential features were:

- quiz_avg

- attendance
- failures
- trend
- studytime

The feature engineering process significantly improved predictive performance by creating meaningful indicators such as attendance, quiz average, and performance trend.

A comparison between the full feature model and the reduced feature model showed that the reduced feature model maintained strong predictive accuracy while improving interpretability and reducing computational complexity. This confirms that carefully selected meaningful features are more valuable than simply increasing the number of input variables.

Further feature selection was assessed by looking at the full feature model and the reduced feature model performances using XGBoost feature importance. The full feature option left all available academic, behavioral, and demographic variables intact, while the reduced feature option included only the most influential predictors concerning quiz average, attendance, failures, and performance trend.

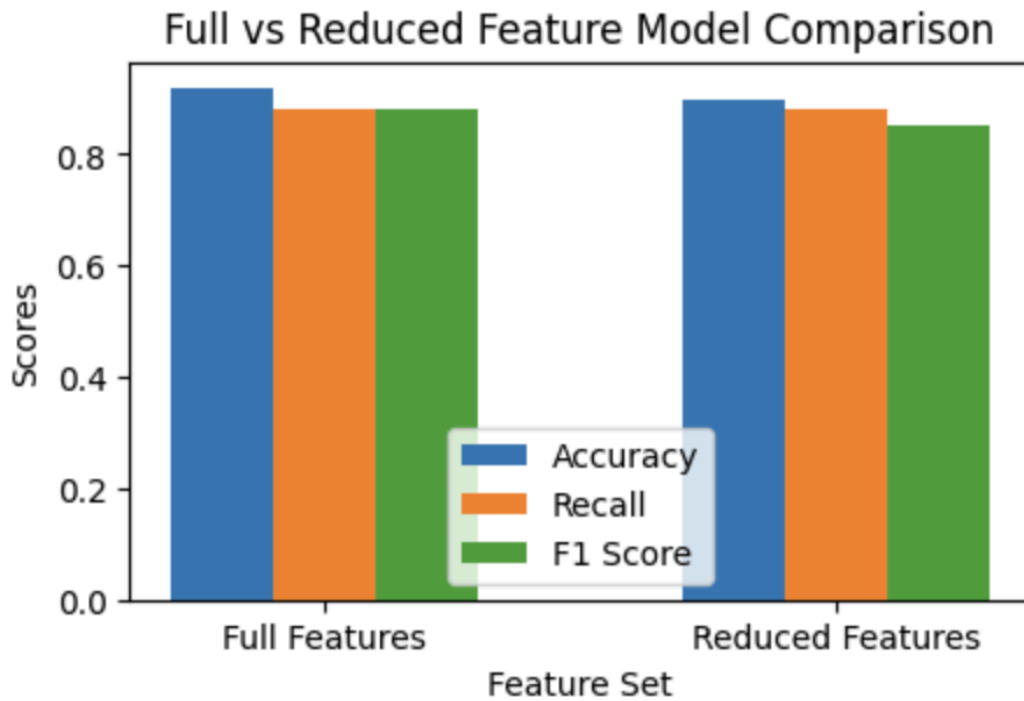


Fig. Feature model comparison

Table 5: Comparison Between Full Feature Model and Reduced Feature Model

Model Version	Accuracy	Precision	Recall	F1 Score	Observation
Full Feature Model	0.92	High	0.88	0.88	Best overall performance
Reduced Feature Model	Slightly Lower	Slightly Lower	Slightly Lower	Slightly Lower	Better interpretability, lower complexity

The comparison showed that the full feature model was slightly better than the reduced feature model for overall prediction accuracy and classification stability. That is to say that, XGBoost is enhanced by more additional features along with their interactions, albeit when some of the variables are less significant individually. Therefore, the full feature model was deemed as the final XGBoost model to use for the proposed framework, despite the interpretable nature and reduced complexity of the reduced feature model, owing to the superiority of predictive performance exhibited by the former.

5.4 Explainability Using SHAP

One major limitation of traditional machine learning systems is the lack of interpretability. To address this issue, SHAP (SHapley Additive Explanations) was used to explain the XGBoost model predictions.

SHAP analysis revealed that quiz average had the strongest influence on academic risk prediction. Students with lower quiz averages were significantly more likely to be classified as high risk. Attendance was the second most important factor, showing that consistent classroom participation strongly supports academic stability. Prior failures also had a major negative effect on prediction outcomes.

This explainability is important because it allows educators to move beyond simple prediction and understand the actual reasons behind student risk. Instead of receiving only a risk label, administrators can identify specific causes and design more effective intervention strategies.

Table 2: Performance Comparison of Classification Models

Feature	Importance	Interpretation
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quiz_avg	Very High	Strongest academic performance indicator
attendance	High	Low attendance increases academic risk
failures	High	Previous failures strongly affect future risk
trend	Medium	Declining performance signals early warning
studytime	Medium	Low study time affects performance stability

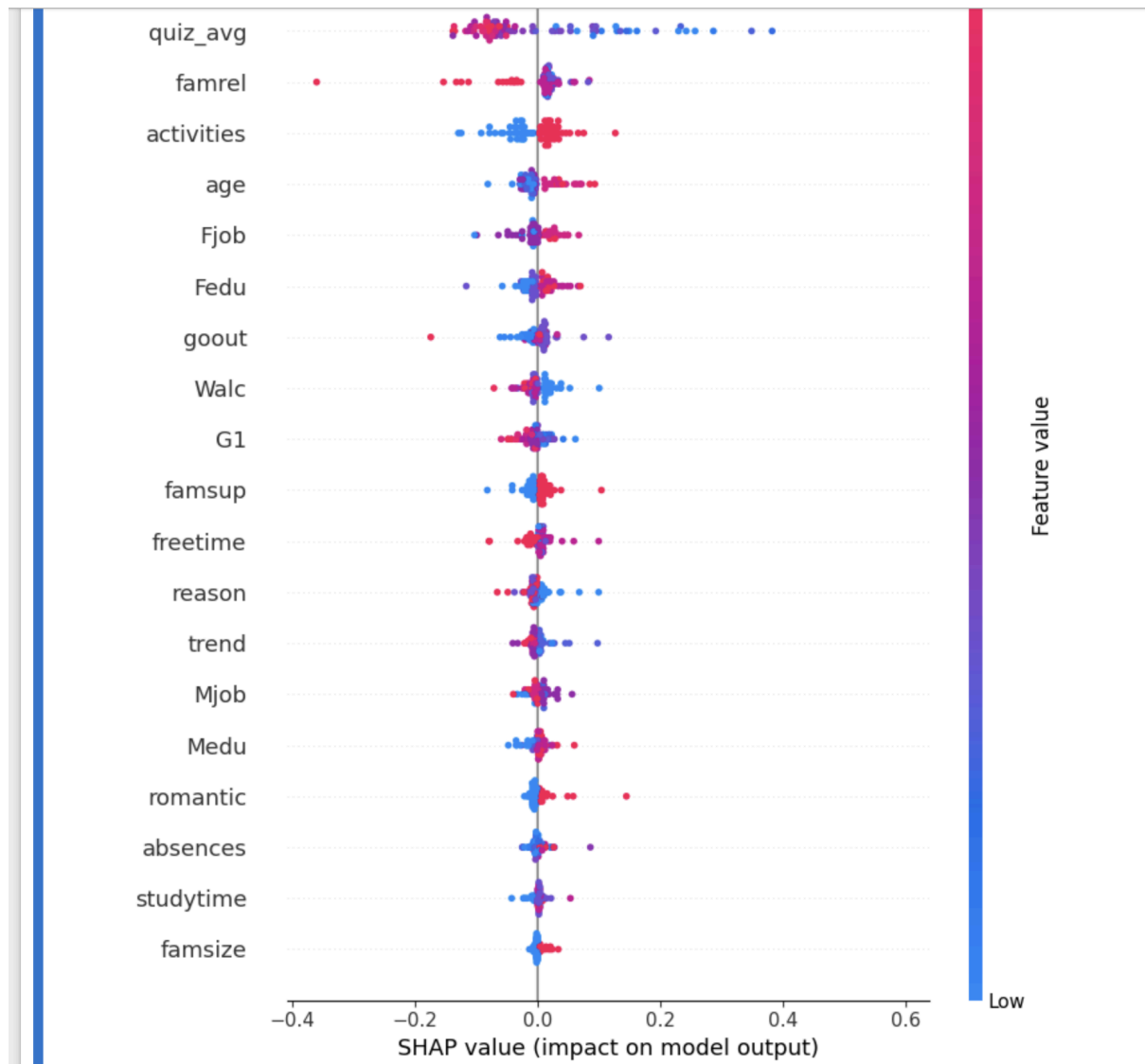


Fig 4: SHAP Feature Importance Plot for XGBoost Model

For example, a student with low attendance but strong quiz performance may require behavioral monitoring, while a student with repeated failures may require direct academic counseling.

5.5 Anomaly Detection Results

Isolation Forest was used to detect anomalous students whose behavior patterns differed significantly from the majority of students. Approximately 10% of the dataset was identified as anomalous.

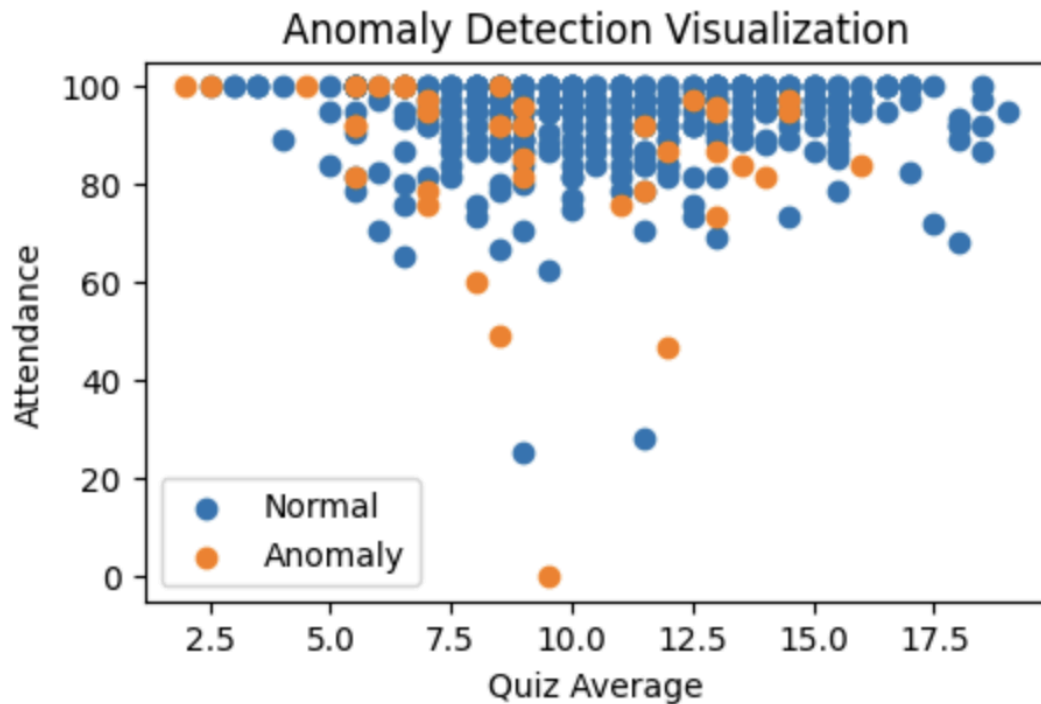


Fig 5: Anomaly Detection Visualization Using Isolation Forest

An important observation was that some anomalous students were not classified as high risk by the supervised classification model. These students appeared academically stable based on grades but showed unusual behavioral indicators such as inconsistent attendance, sudden performance decline, or irregular engagement patterns.

This finding highlights the importance of anomaly detection as an additional early warning mechanism. Traditional classification alone may fail to detect such hidden-risk students, while anomaly detection provides a second layer of protection.

This improves the practical reliability of the proposed framework by enabling preventive intervention before visible academic decline occurs.

5.6 Rule-Based Intervention Outcomes

The final stage of the system was the rule-based intervention framework, which combined risk prediction labels and anomaly detection results to generate personalized academic recommendations.

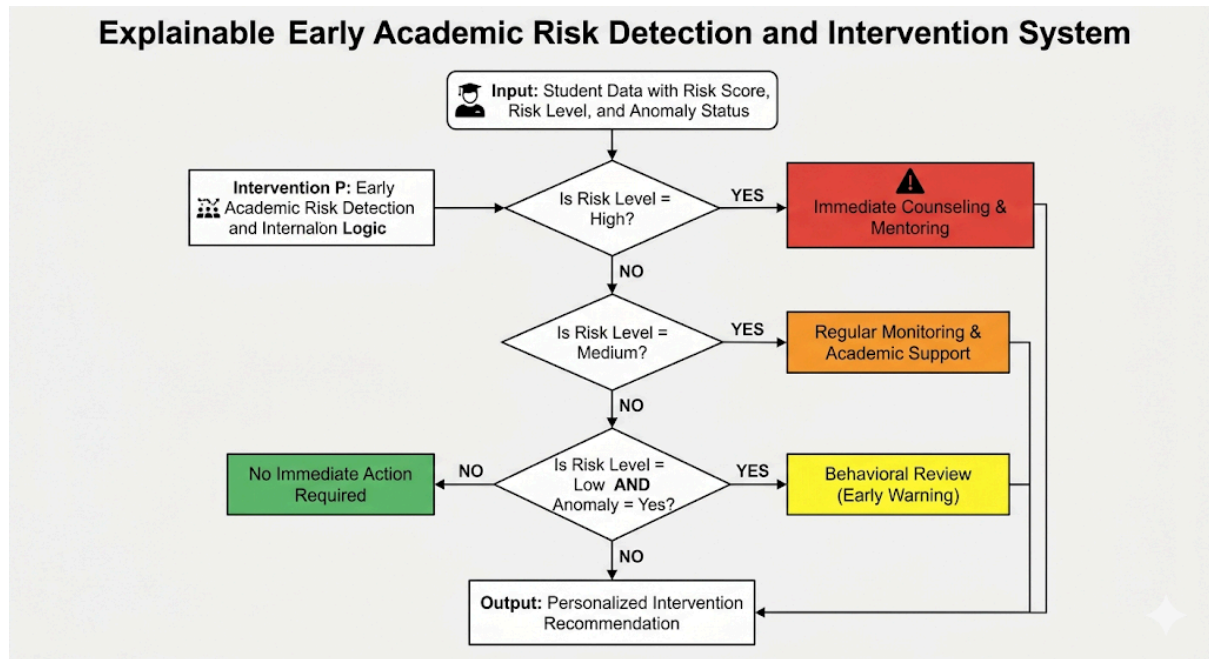


Fig 6: Rule-Based Intervention Framework for Academic Risk Management

Students predicted as high risk and anomalous received stronger intervention recommendations such as:

- Immediate academic counseling
- Weekly performance monitoring
- Personalized mentoring sessions
- Behavioral review and family consultation

Students classified as low risk but anomalous were assigned preventive monitoring and behavioral observation to avoid future decline.

This intervention layer transforms the model from a purely predictive system into an actionable educational support framework. The goal is not only to predict failure but to prevent it.

5.7 Discussion

The results confirm that combining classification, explainability, anomaly detection, and intervention planning creates a stronger educational prediction system than traditional classification-only approaches.

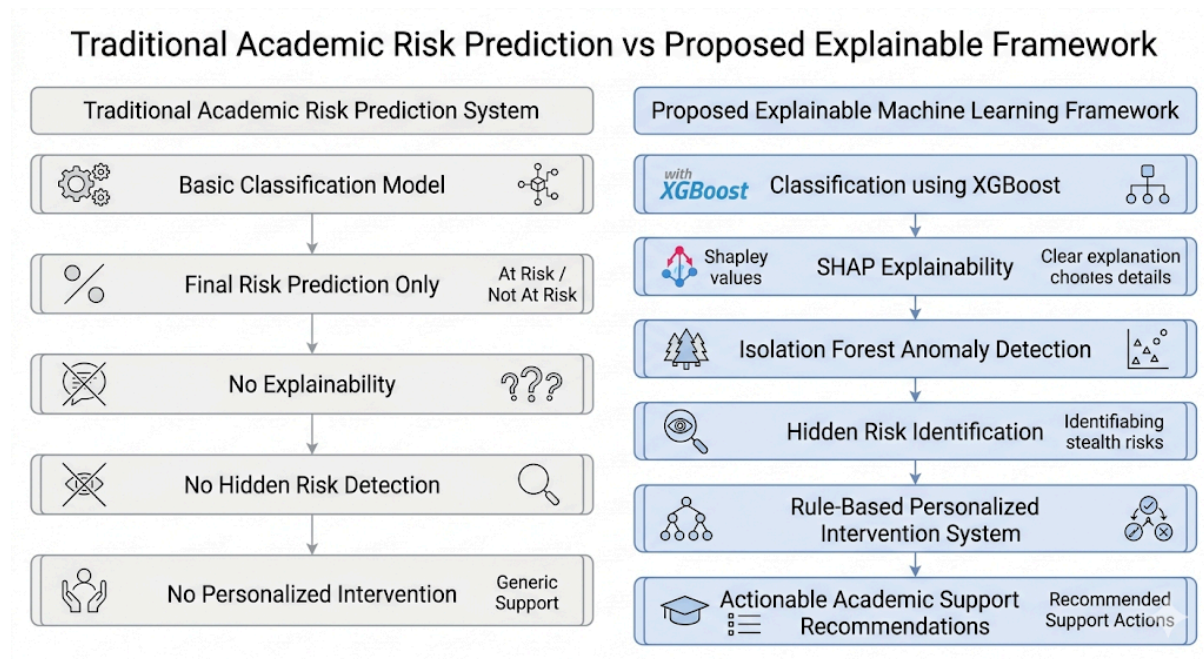


Fig 7: Comparison Between Traditional Prediction Systems and the Proposed Explainable Framework

Unlike many previous studies that focus only on final accuracy, this framework improves practical usability by answering three important questions:

1. Which students are at risk?
2. Why are they at risk?
3. What action should be taken next?

This makes the system more suitable for institutional deployment where transparency and intervention planning are as important as prediction performance.

The integration of XGBoost, SHAP, and Isolation Forest provides both strong predictive accuracy and real-world decision support, making the proposed framework a significant improvement over conventional student performance prediction models.

6. CONCLUSION

This research presented an Explainable Machine Learning Framework for Early Academic Risk Detection and Intervention designed to identify students at risk of poor academic performance before final academic failure occurs. The proposed system combines supervised classification, explainable artificial intelligence, anomaly detection, and a rule-based intervention mechanism to improve both prediction accuracy and practical educational decision-making.

The study used the UCI Student Performance Dataset containing academic, behavioral, and socio-demographic information of students. Feature engineering was performed to create meaningful indicators such as attendance, quiz average, and performance trend, which improved both model performance and interpretability. Three classification models, namely Logistic Regression, Decision Tree, and XGBoost, were evaluated for academic risk prediction.

Among the tested models, XGBoost achieved the best performance with an accuracy of 92%, recall of 88%, and F1-score of 0.88. The confusion matrix further confirmed that the model maintained a strong balance between correctly identifying high-risk students and minimizing false predictions, making it suitable for real-world educational use.

To overcome the limitations of black-box prediction systems, SHAP was used to provide model explainability. The analysis revealed that quiz average, attendance, and prior failures were the most influential factors affecting student academic risk. This helped transform the model from a simple prediction tool into an interpretable decision support system for teachers and academic administrators.

Isolation Forest was applied as an anomaly detection technique to identify hidden-risk students who were not classified as high risk by traditional supervised models.

Approximately 10% of students were detected as anomalous, showing unusual behavioral patterns despite acceptable academic performance. This provided an additional early warning layer for proactive intervention.

A rule-based intervention system was developed using risk labels and anomaly detection results to generate personalized recommendations such as counseling, monitoring, mentoring,

and behavioral review. This made the framework practically useful by connecting prediction results directly to actionable academic support strategies.

The major contribution of this research is the development of an integrated framework that answers not only who is at risk, but also why they are at risk and what actions should be taken to improve their academic outcomes. This makes the system more effective than traditional classification-only approaches.

The comparison of the full feature model and the reduced feature model added further evidence of the framework's reliability. Even though the reduced feature model enhanced interpretability and computation efficiency, the full feature model had a slight advantage in terms of predictive performance. Therefore, the inclusion of additional academic, behavioral, and demographic features offers meaningful interactions that enhance accuracy in predicting risk, thus making the full feature model more appropriate to be used in the final deployment.

For future work, additional indicators such as psychological factors, financial background, emotional well-being, and real-time Learning Management System (LMS) interaction data can be incorporated to further improve prediction quality. Deep learning models and more advanced intervention recommendation systems may also be explored to enhance scalability and institutional adoption.

In conclusion, the proposed framework demonstrates that combining machine learning, explainability, anomaly detection, and intervention planning can significantly improve early academic risk detection and support more effective student retention strategies in modern educational environments.

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